

Grounded Intelligence: Scaling Multimodal Medical Retrieval and Verifying LLMs



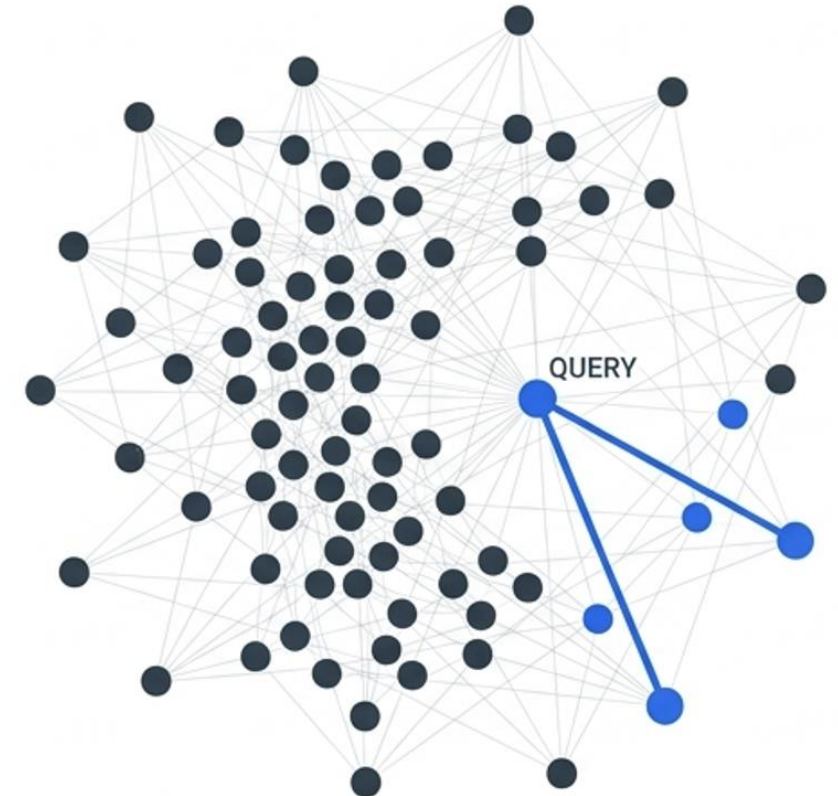
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Questions at the beginning

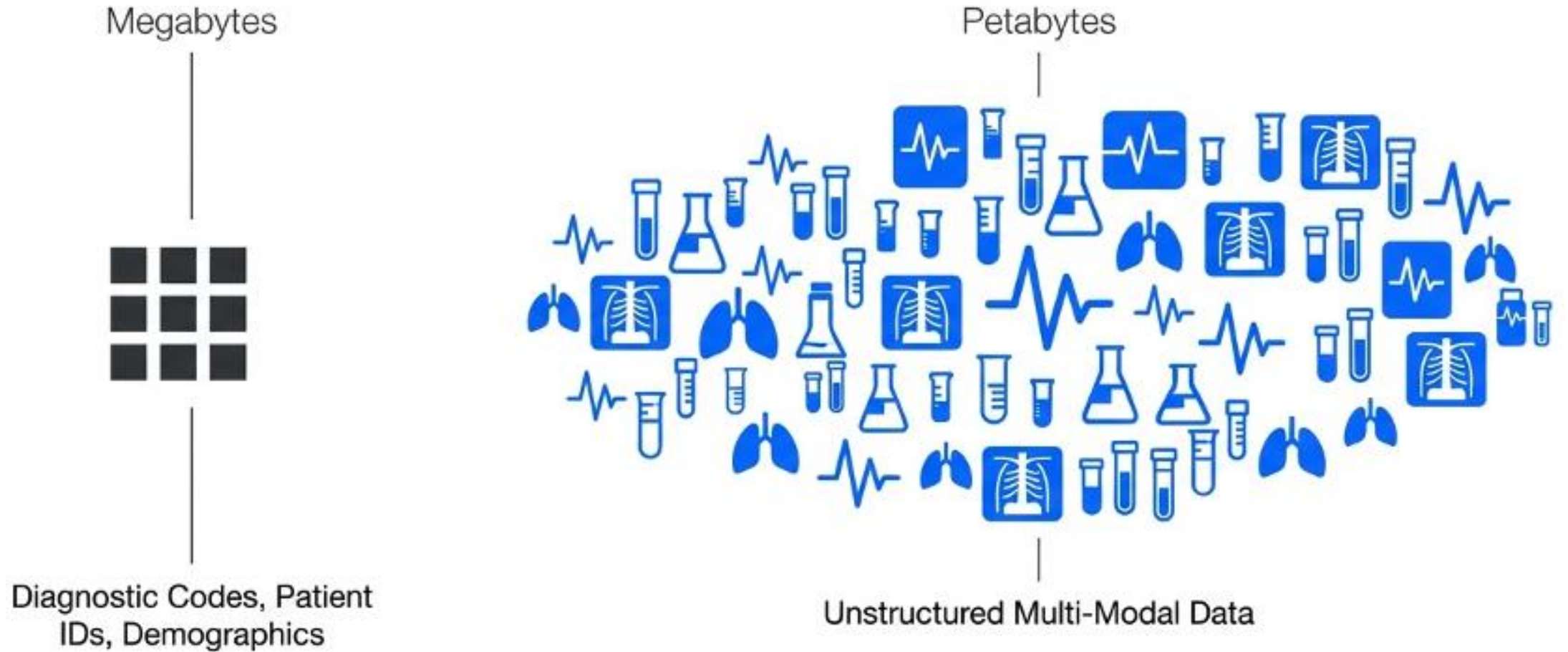
- How can we scale medical similarity search to billion-scale records without sacrificing speed or accuracy?
- How can we prevent and detect LLM hallucinations in high-stakes fields like medicine?
- How can we bridge disparate medical modalities to discover complex clinical paths?

Part 1: The Foundation of Billion-Scale Search

- Modern machine learning represents images, videos, and text as high-dimensional vectors.
- Finding exact nearest neighbors (k-NN) across a database of one billion vectors, making exhaustive search computationally impractical
- Approximate search (IVFADC) aggressively optimized for micro-architecture of GPU / CPU hardware
- IVFADC combines coarse clustering (IVF) with compressed vector representations (PQ) and asymmetric distance computation (ADC) to enable fast approximate nearest-neighbor search over billion-scale vector databases



Exhaustive Search Bottleneck



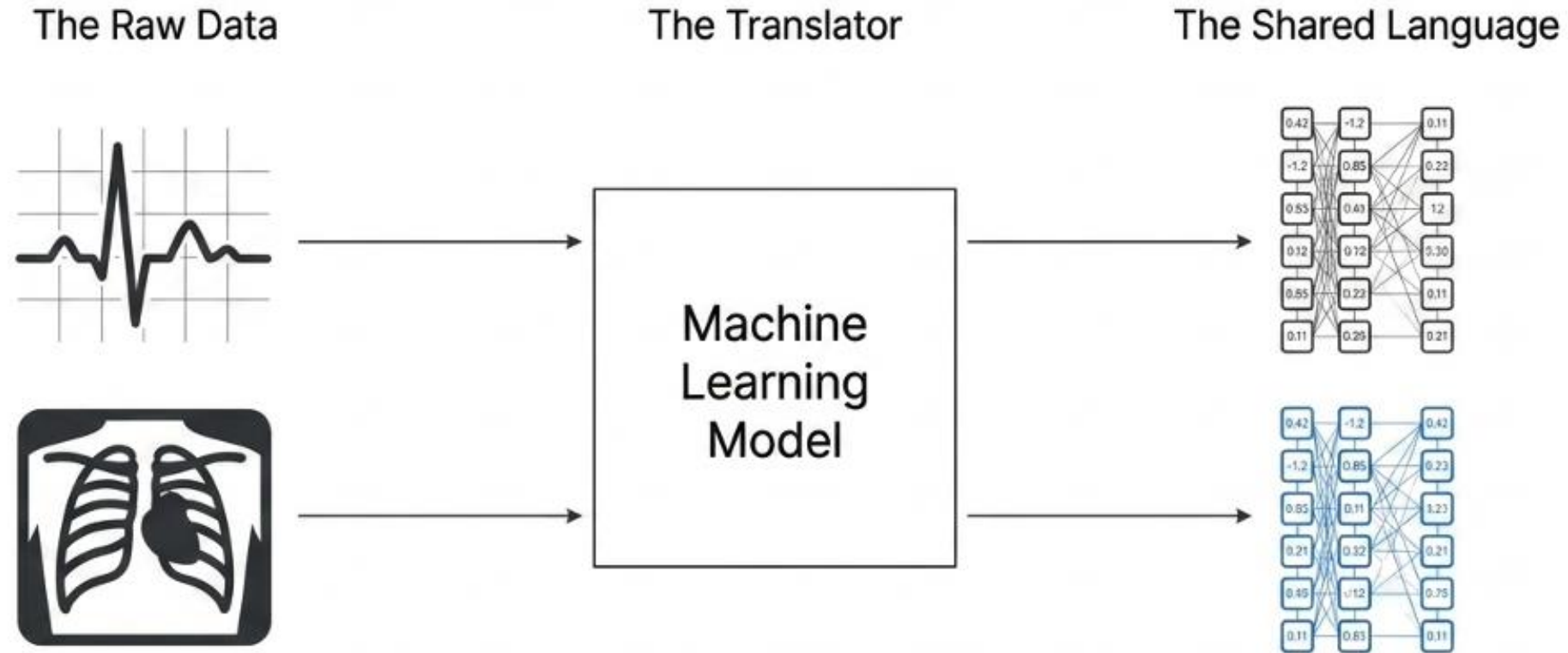
Exhaustive search clashes when traversing billions of high-dimensional clinical records

Exact match fails to capture underlying relationships across different health record modalities

Rethinking Retrieval: Structured Matching versus Numerical Similarity

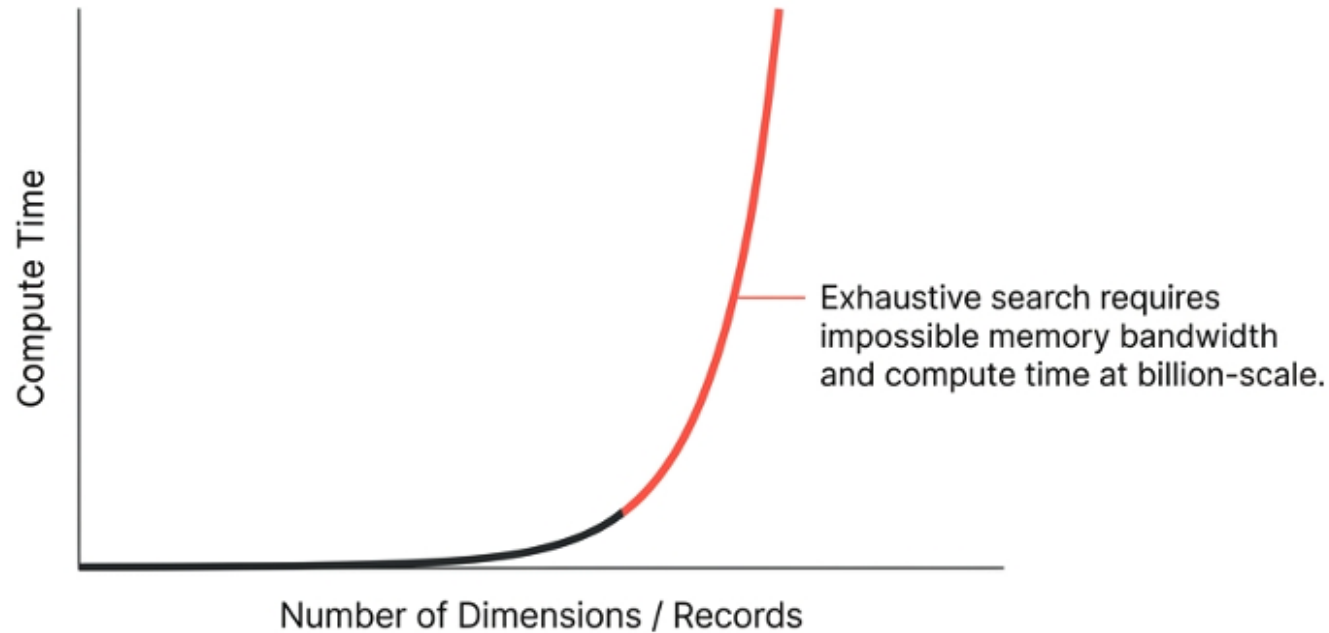
Structured Matching (Traditional Retrieval)	Numerical Similarity (Topology Search)
Mechanism: Row / Column Database Queries	Mechanism: High-Dimensional Vector Space
Criteria: Exact Patient IDs & rigid ICD codes	Criteria: Geometric Distance (L2) between records
Result Type: Binary (Match / Not Match)	Result Type: Smooth transitions (k-Nearest Neighbors)
Limitation: Blind to cross-modality clinical context	Advantage: Discovers highly relevant neighboring cases regardless of tags

Translating Modalities into a Shared Manifold



Embeddings: information is encoded into a shared latent space by machine learning

The Curse of Dimensionality

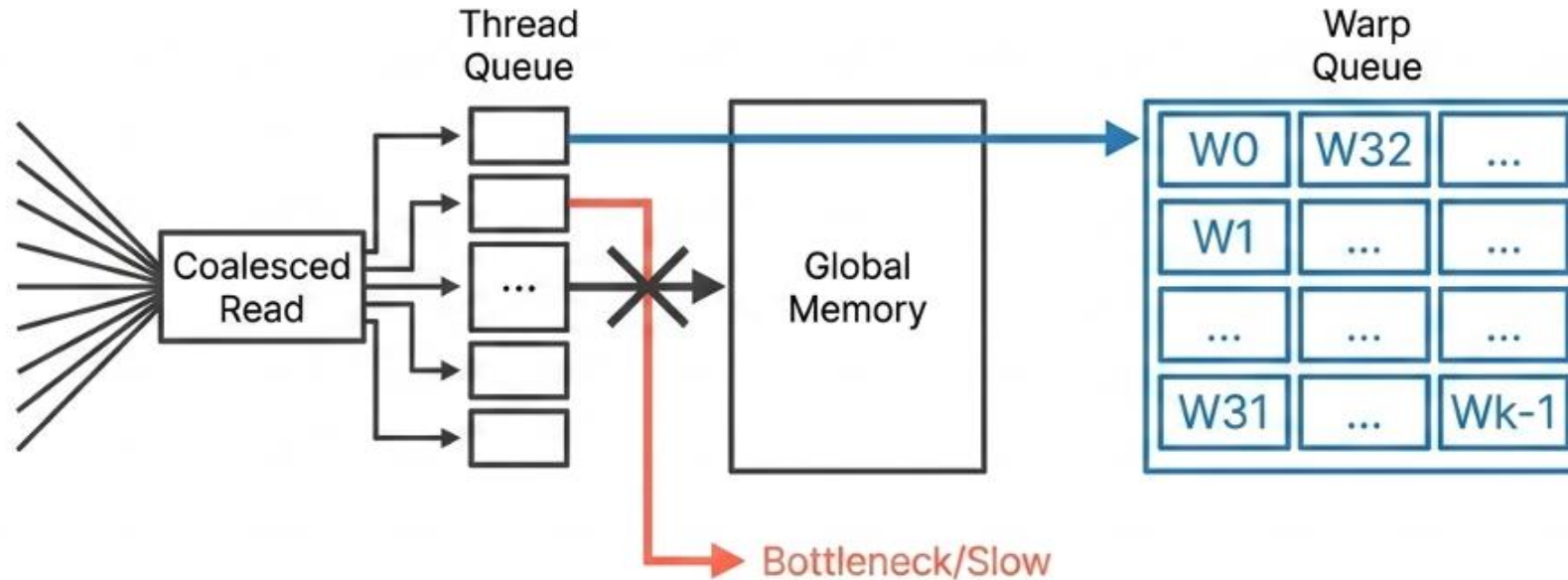


To query 1 billion vectors effectively, we must abandon CPU-bound operations.



The cure is highly parallel computing such as GPU architecture: extreme data compression (product quantization) and bypassing slow memory hierarchies (WarpSelect)

Bypass the Bottleneck: WarpSelect

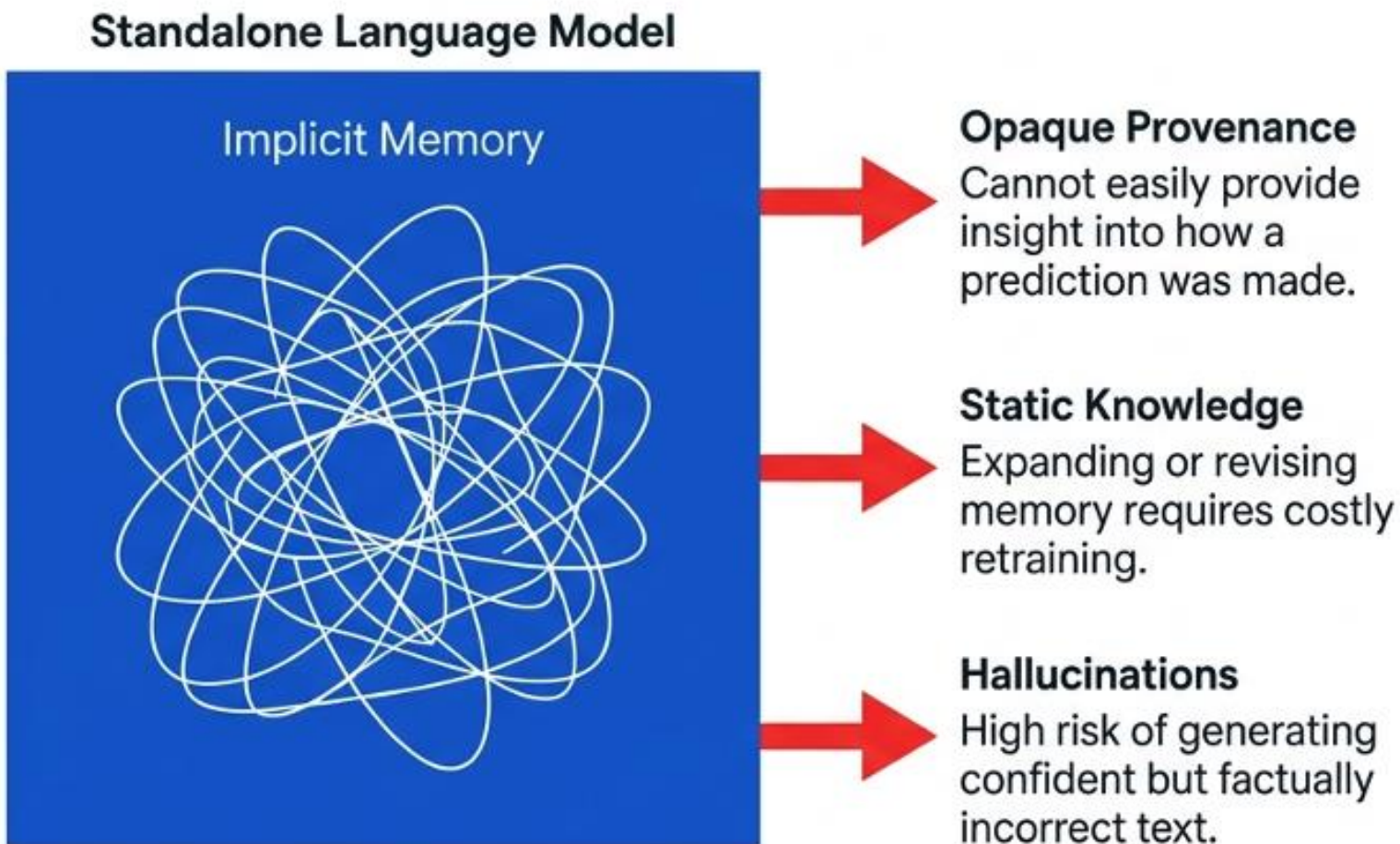


- **Coalesced Read** → GPU threads load candidate distances efficiently from global memory.
- **Thread Queue** → Each thread locally stores its best candidates in registers.
- **Warp Queue** → Candidate lists from 32 threads are merged cooperatively into a warp-wide top- k queue.
- **Global Memory** → Only the final selected candidates are written back, minimizing slow memory operations.
- **Bottleneck** → Global memory remains the primary performance bottleneck, while WarpSelect minimizes its impact by keeping almost all selection operations inside the warp.

Part 2: Retrieval-Augmented Generation (RAG)

Limit of Implicit Memory

Pre-trained LLMs act as parameterized implicit knowledge bases, creating critical bottlenecks for factual accuracy



The Hybrid Solution: feeding LLM with a differentiable access mechanism to an explicit external memory

Non-Parametric Memory

What it is: A dense vector index of Wikipedia.

Mechanism: Accessed via a pre-trained neural retriever (DPR).

Scale: 21 million 100-word chunks of text.

Trait: Explicit, inspectable, and easily updatable.

Parametric Memory

What it is: A pre-trained seq2seq transformer.

Mechanism: A Generator (BART-large) that synthesizes retrieved data.

Scale: 400 million trainable parameters.

Trait: Implicit, specialized in natural language generation and reasoning.

Retrieval-Augmented Generation (RAG)

Standalone Generator VS External Document Index

Feature	Parametric Memory (BART)	Non-Parametric Memory (DPR)
Storage Mechanism	Implicitly encoded in model weights	Explicit text passages in a dense vector index
Updateability	Requires expensive retraining	Instant (Add, remove, or swap documents)
Interpretability	Opaque (Black box)	Transparent (Read the exact retrieved document)
Size/Footprint	Fixed (400M parameters)	Scalable (15.3 billion vector values)
Function	Synthesizing language and logic	Fact-checking and knowledge retrieval

Two Generation Approaches

RAG-Sequence Model

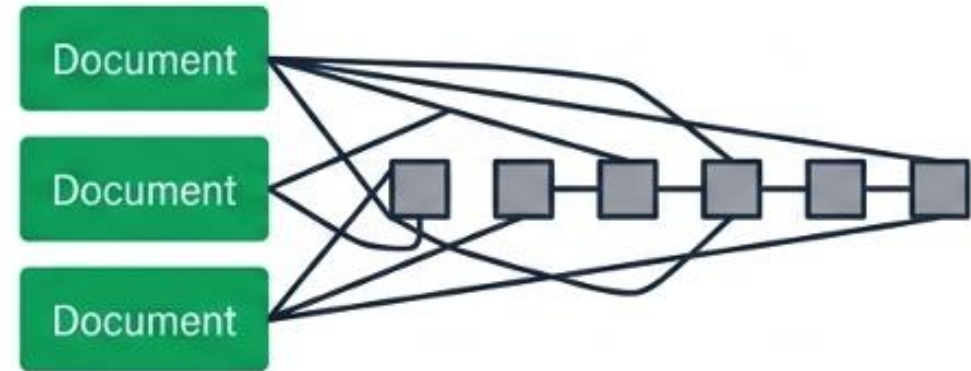
One document per answer.



The model uses the same retrieved document to predict every target token in the complete sequence. The probability is marginalized at the sequence level.

RAG-Token Model

Multiple documents per answer.



The model can draw a different latent document for each target token. The generator chooses content from several documents when producing a complex compound answer.

Depending on the complexity of the task, RAG marginalize over latent documents in two distinct ways.

Eradicating LLM Hallucinations

Human Assessment on Factuality



Task: Define 'middle ear'

Standalone BART



The middle ear is the part of the ear between the middle ear and the nose.

RAG-Sequence

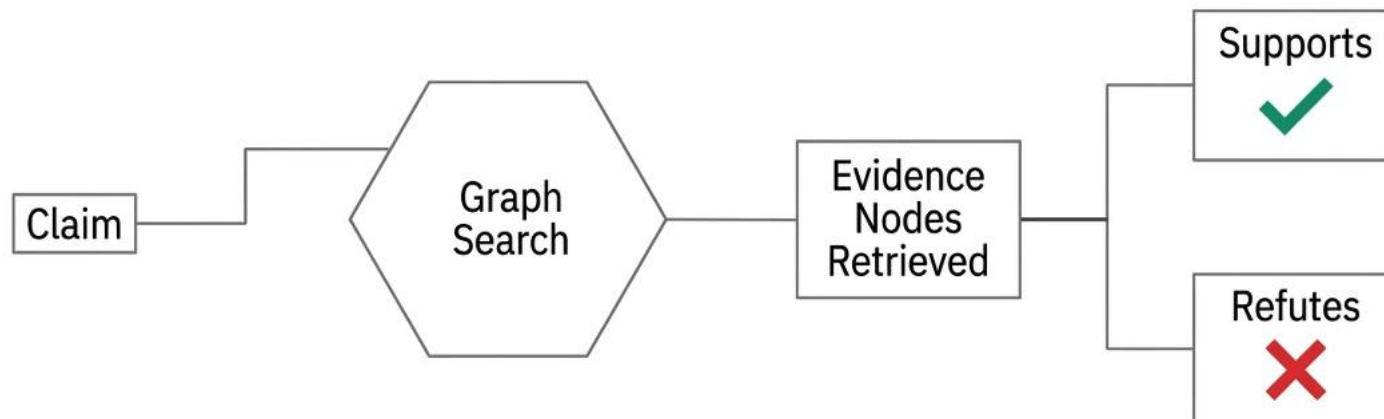
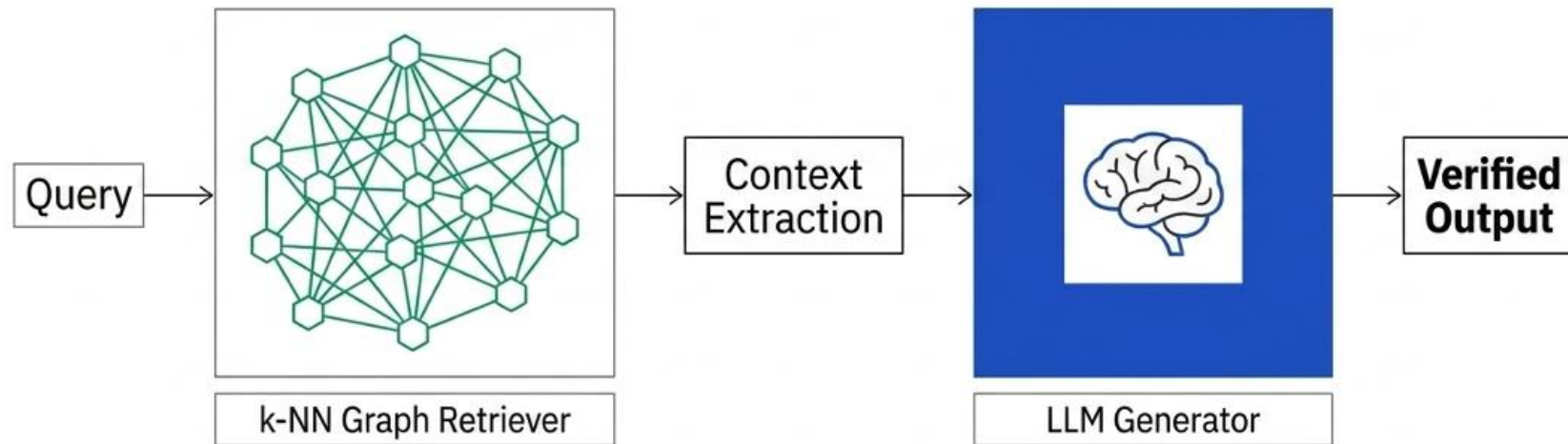


The middle ear includes the tympanic cavity and the three ossicles.

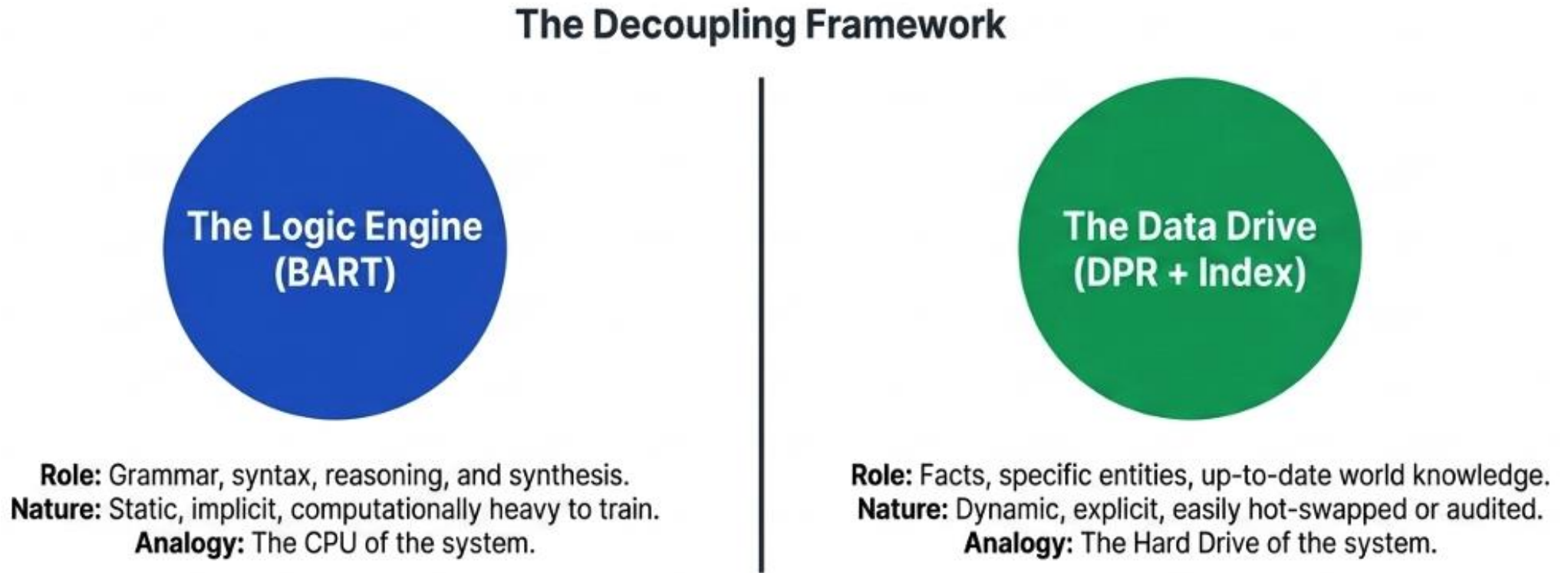
In the test for highly specific jeopardy questions, RAG dramatically outperforms standalone LLM in factuality

The Synthesis Architecture with Knowledge Graph

Combining the semantic reasoning of parametric LLMs with targeted RAG retrieval, the KG forces the LLM to generate answers grounded entirely in verified, multimodal pathways.



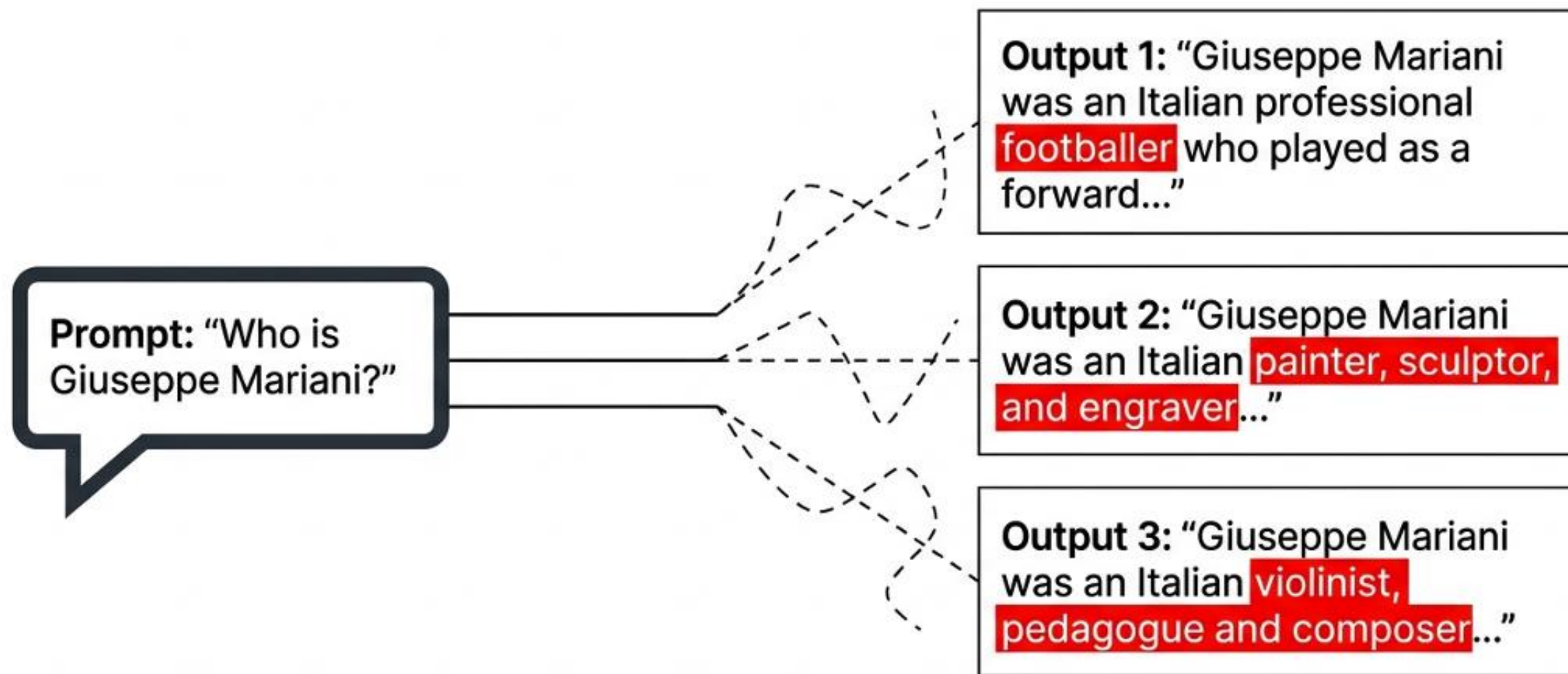
Dynamic Intelligence



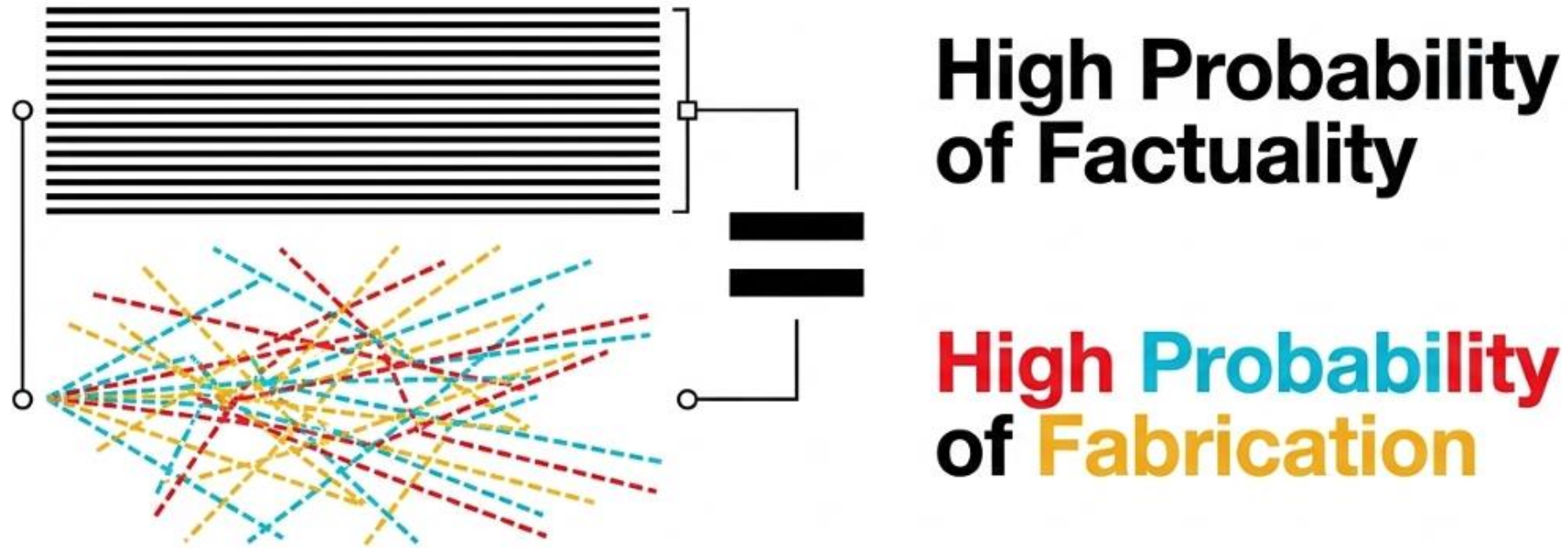
By separating the ability to speak from the knowledge reservoirs, RAG create AI systems to minimize hallucination, cite their sources, and evolve instantly according to changes.

Part 3: LLM Hallucination

LLM hallucination is the fluent, confident generation of realistic but entirely fictitious information. It invalidates LLM outputs for high-stakes applications

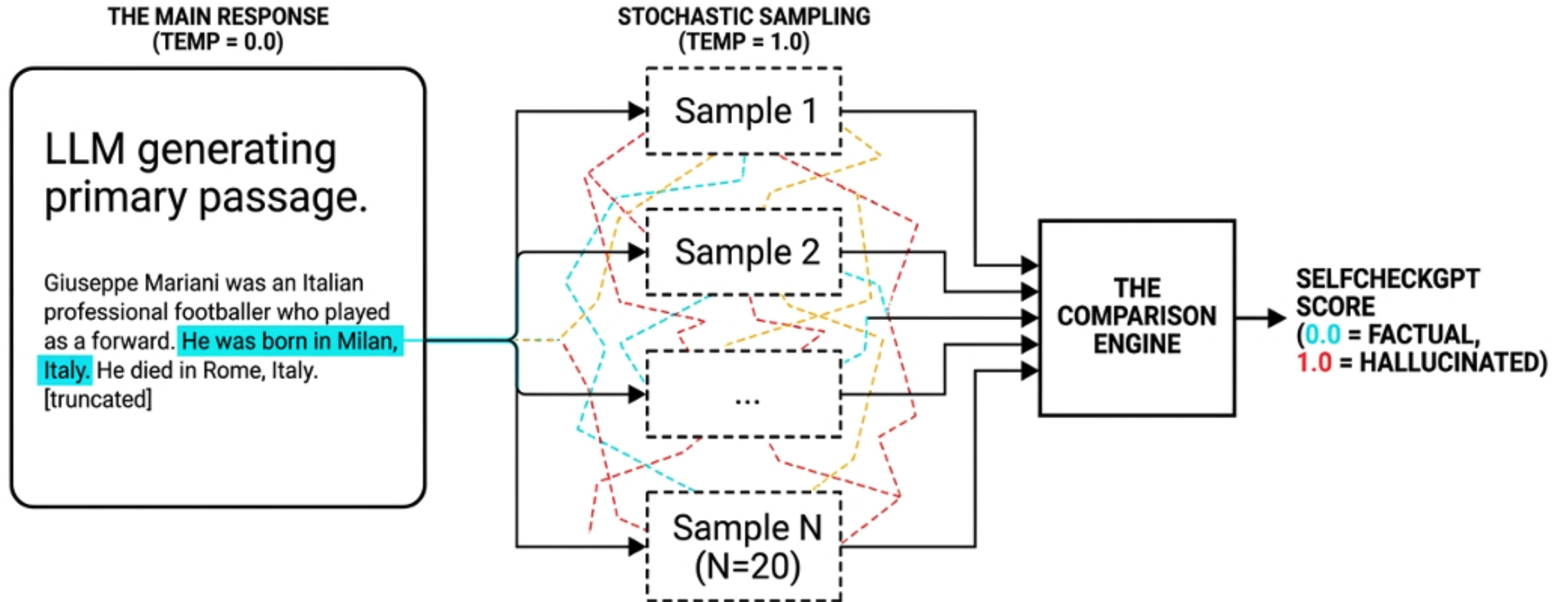


If an LLM knows a concept, its stochastically sampled responses will be similar and consistent. If an LLM is hallucinating, its samples will diverge and contradict.

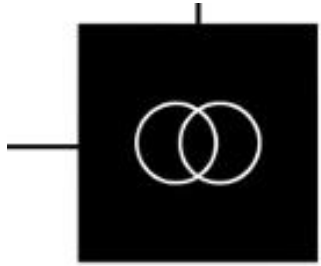


Information consistency = Factuality

SelfCheckGPT

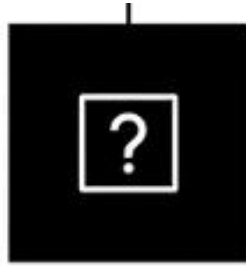


Five Ways to Measure Consistency



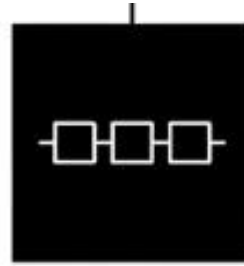
1. BERTScore

Measures average semantic similarity between the main sentence and the samples.



2. Question-Answering (QA)

Generates multiple-choice questions from the main response and checks if samples yield the same answers.



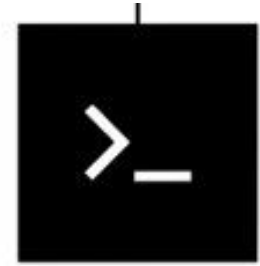
3. N-gram

Trains a simple local language model on the samples to check the probability of main response tokens.



4. Natural Language Inference (NLI)

Calculates a Contradiction Score between the samples (premise) and main response (hypothesis).

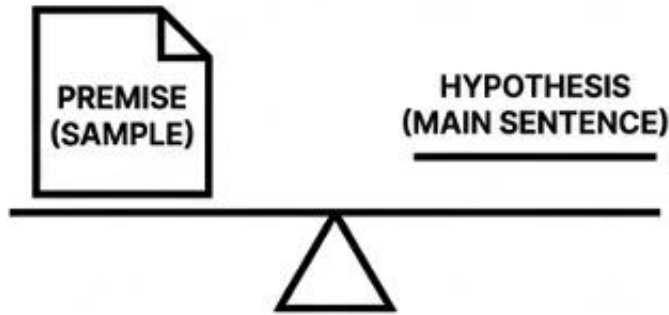


5. LLM Prompting

Directly asks an LLM if the sample supports the sentence.

NLI VS UNIGRAM: some variants

SELFHECK-NLI (NATURAL LANGUAGE INFERENCE)



Uses an NLI model (like DeBERTa-v3) to classify the relationship. Only 'entailment' and 'contradiction' logits are evaluated.

ADVANTAGE: **High performance** without secondary API costs. The best trade-off of accuracy and compute.

SELFHECK-UNIGRAM



Finds the token in the main sentence with the lowest occurrence probability across all N samples.

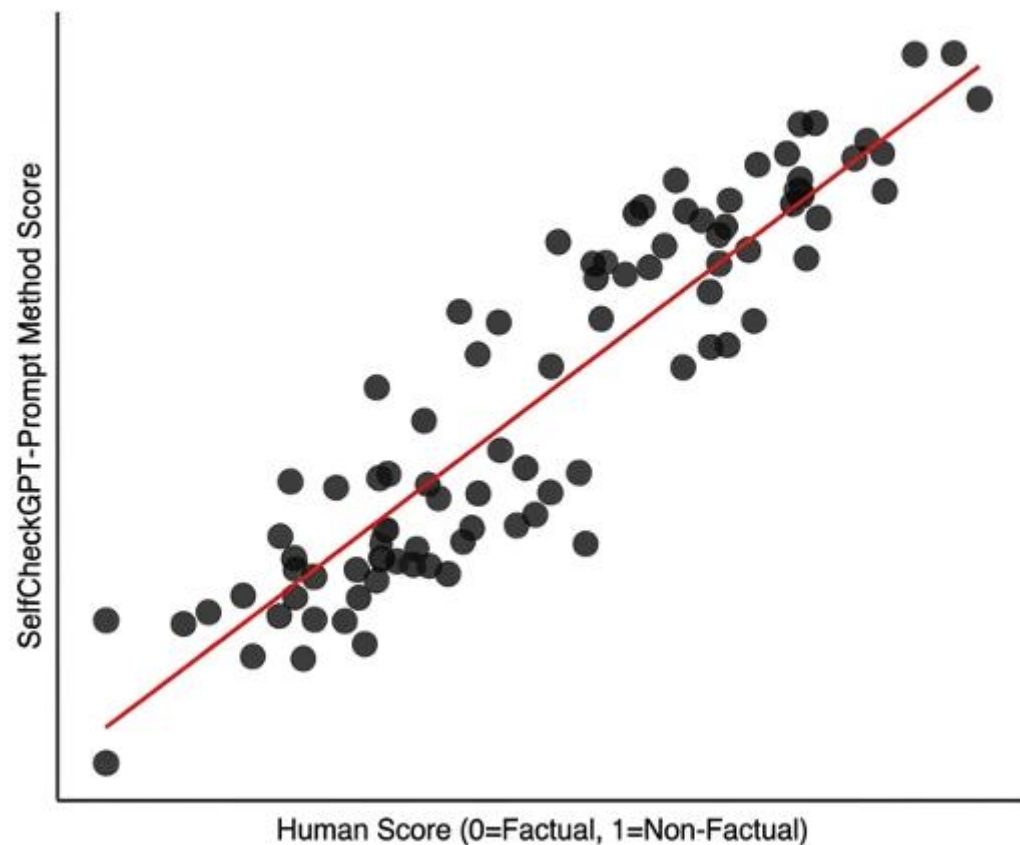
ADVANTAGE: The **absolute fastest** and **computationally cheapest** method. Highly effective at spotting rare, fabricated terms.

Sentences-Level Detection



KEY INSIGHT: SelfCheckGPT decisively outperforms Grey-Box methods. You do not need access to internal model states to beat internal model metrics.

Passage-Level Ranking



78.32

Pearson correlation with human judgment.

TAKEAWAY: The framework accurately scores entire documents, making it highly viable for automated QA pipelines reviewing long-form AI generation.

Can LLM check its own work?

GPT-3 evaluating GPT-3	ChatGPT evaluating GPT-3
Cost per 1k tokens: \$0.020	Cost per 1k tokens: \$0.002 (10x cheaper)
Performance (Pearson): 73.11	Performance (Pearson): 76.47

Takeaway: Yes, models can check their own work. However, using a **chat-optimized, cheaper model** (ChatGPT) as the evaluator actually yields a **slight performance edge** and **massive cost savings** over the raw base model.

Findings

Information consistency is a reliable proxy for factuality.

Hallucinations are mathematically measurable through stochastic divergence.

Black-box outperforms Grey-box.

You do not need white-box access, internal model states, or token probabilities to detect AI hallucinations effectively.

Zero-resource evaluation is viable.

SelfCheckGPT eliminates the strict dependency on costly, brittle external databases for general fact-checking pipelines.